



Alexa Blueprints

Amazon has introduced 'Alexa Blueprints' that lets users create their own customized Alexa skills or responses. <https://dgit.in/blueprints>



AI security camera

The new Lighthouse security camera, priced at \$300, uses AI technology to identify you, your family and pets. <https://dgit.in/aisecurity>

WESTWORLD

The use of CGI in mainstream media got its first real boost with *Westworld*. We're talking about the original movie here. To generate the viewpoint of the gunslinger, the main antagonist, the filmmakers used an FR80 system. They had to record the scenes that needed to be simulated in the gunslingers vision, feed it to the FR80 and process it on the onboard computer. It involved colour separation, repeated scans and actors having to wear contrasting clothing to be distinguishable in the output footage. You can see why it was considered groundbreaking at the time: <https://dgit.in/GSlngerVsn>.

FUTUREWORLD

Westworld inspired a sequel, *Futureworld*, which was notable for using 3D animated graphics for the first time. A few seconds of footage showed a 3D animated hand and head. Part of this footage was taken from *A Computer Animated Hand*, a 1972 American computer-animated film.

STAR WARS

George Lucas jumped into the CGI fray with *Star Wars*. Larry Cuba, a computer-animation artist, got the contract to develop the CGI wireframe animation of a wireframe Death Star for the Rebel Alliance briefing scene. Cuba produced the sequences using the Grass Programming Language and a Vector General 3D connected to a PDP-11/45 system at the the University of Illinois.

TRON

The movie *Tron* was significant in the world of CGI for multiple reasons. One was its extensive use of CGI and computer animation. The other was the fact that this was probably the last major CGI project in Hollywood that did not involve Lucasfilm for a really long time.

The movie included 15-20 minutes of computer animated footage. Interestingly, the final work was done on a computer with only 2MB of memory and 330 MB of storage, which led to designers blackening things out

beyond a certain distance. "When in doubt, black it out" became the motto of the Computer Effects department.

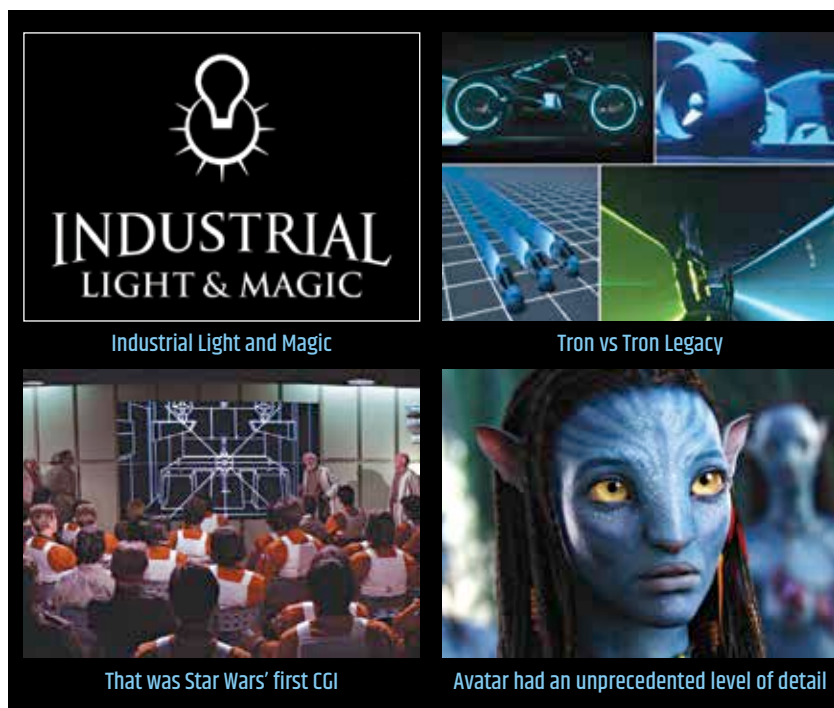
INDUSTRIAL LIGHT AND MAGIC

Industrial Light and Magic (ILM) was created by George Lucas to be the CGI arm of Lucasfilm back in 1975. ILM's filmography will put anyone in Hollywood, or maybe film industries across the world, to shame. Starting from *Star Wars: Episode IV – A New Hope*, the studio has worked on numerous titles in the *Star Trek*, *Indiana Jones*, *Jurassic Park*, *Godfather*, *Terminator*, *Jumanji* franchises as well as iconic individual

of CGI achievements with *Toy Story*, *Titanic*, the *Matrix* and more. It was truly a wonderful time for CGI.

PHOTOREALISM AND MOTION CAPTURE

The turn of the millennium brought a newfound focus on making CGI more believable. Dr Aki Ross, the heroine of Hironobu Sakaguchi's *Final Fantasy: The Spirits Within*, predated Gollum and the Train conductor from *Polar Express* in terms of rendering detail. The production process was very expensive and the 141,964 frames involving her required 90 minutes to



Industrial Light and Magic

Tron vs Tron Legacy

That was Star Wars' first CGI

Avatar had an unprecedented level of detail

titles like *Forrest Gump*, *Saving Private Ryan* and more. They've even worked on recent titles like the *Harry Potter* series, *Pirates of the Caribbean*, *Avatar* and MCU movies.

Just like the filmography, the list of milestones achieved by ILM since its inception leaves no doubt towards who is responsible for the current state of CGI. *Jurassic Park* in '93, was the first time digital technology was used to create a complete and detailed living creature. A marvelous blend of animatronics and bleeding edge CGI. The rest of the 90s saw a smorgasbord

render - generating a final result of 15 terabytes of animation.

In the next couple of years, Peter Jackson's work with Gollum in *LotR* and Robert Zemeckis' work with the train conductor on the *Polar Express* would bring mo-cap marker based motion capture performances to the forefront. James Cameron was pushing the boundaries of CGI with *Avatar*. Consider this – each Pandoran plant consisted of a million CG polygons, whereas Gollum was 50 polygons. It set the standard for facial expression capture for years to come. 